List 5- Fin. Ele

1. (**Linear Algebra**) Show that of if $A \in \mathbb{R}^{n\times n}$ is positive definite ($V^\intercal AV > 0\ \forall V \in \mathbb{R}^n$), then, A is invertible.

Suppose by absurdity that $\exists w \in \mathbb{R}^n$ s.t. $Aw = 0\cdot w$, that is, $\lambda = 0$ is an eigenvalue and w is the respective eigenvector, with $w \neq 0$

$Aw = 0 \Rightarrow w^\intercal Aw = w^\intercal 0 = 0$

But $w^\intercal Aw > 0$. Absurd. So A doesn't have a 0 eigenvalue $\Rightarrow$ A is invertible

2. In the optimallity of the Galerkin approximation u_h, it is used that in the energy norm $\|\cdot\|_a$, the coercivity and continuity constants are both equal to 1. Explain.

Coercity: $a(u,u) > \alpha \|u\|_k^2$ (1)

Continuity: $|a(u,v)| \leq N_a \|u\|_k \|v\|_k$ (2)

Taking $\|\cdot\| = \|\cdot\|_a$

$|a(u,v)| \leq \|u\|_a \|v\|_a \Rightarrow N_a = 1$

(Cauchy Schwarz)

$a(u,u) = \|u\|_a^2 \geq \|u\|_a^2 \Rightarrow \alpha = 1$

So the coercity and continuity constants are 1 for the energy norm. This means that, in the Céa's lemma, we have

$$\|u - u_h\|_a \leq \|u - v_h\|_a, \quad \forall v_h \in V_h$$